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Phenomenology, Cosmology, and Physical Applications

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ABSTRACT

This volume assembles the corpus's physical applications—quantum, particle, atomic, gravitational, cosmological, astrophysical, and informational—under a single evidential discipline. Exact reproductions, retrospective comparisons, forecasts, null results, and open mechanisms are reported separately.

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The contract for physical application

1.1 From formal structure to measured physics

The preceding volumes established a language for distinctions, a carrier geometry, and a disciplined account of traversal and readout. None of those achievements, by itself, identifies an electron, a galaxy, or a cosmological parameter. A physical application requires a realisation map

$$\mathcal{R} : \mathcal{S}_{\text{formal}} \longrightarrow \mathcal{S}_{\text{physical}} \quad (1.1)$$

and a measurement map

$$\mathcal{M} : \mathcal{S}_{\text{physical}} \longrightarrow \mathcal{D} \quad (1.2)$$

to calibrated data. An algebraic identity may be exact in $\mathcal{S}_{\text{formal}}$ while its proposed image under $\mathcal{R}$ is false. Numerical proximity cannot supply the missing map.

This volume therefore uses five evidential types.

TABLE 1.1. Application-level evidential types.

Type	Meaning
Established	Published mathematics or experiment, with its actual hypotheses and uncertainty.
Internal result	Reproducible consequence of the corpus's declared definitions or data.
Conditional identification	Consequence obtained only after a stated formal object is mapped to a physical one.
Forecast	A predeclared numerical or qualitative output that later data can test.
Exposure	A missing derivation, non-identifiability, failed route, or observation that can defeat the proposal.

1.2 No scientific freezes

Earlier corpus documents sometimes used *freeze* to mean that a formula, convention, or scoring rule should not be altered while a particular audit was running. That is a valid preregistration device, but it cannot confer truth. In this edition every claim remains revisable. A later version may supersede an equation, mapping, or grade when a proof fails, a convention changes, or evidence improves. The earlier state remains in the provenance ledger so that the change is inspectable.

Words such as “exact”, “forced”, and “parameter-free” always have a domain. “Exact” means exact under the displayed definitions; “forced” names the assumptions that do the forcing; and “parameter-free” does not mean assumption-free, convention-free, or selected independently of the comparison.

1.3 The application sequence

The order is deliberately chronological in logical dependence. Matter representations come first because later atomic and cosmological identifications presuppose a particle language. Atomic structure follows as a finite, controlled spectral problem. Classical theories are then treated as limits and translations. Cosmology comes last because it combines geometry, matter, radiative transfer, inference, and survey selection in one measurement chain.

Matter representations and the Standard Model contact

2.1 What is inherited from established representation theory

The carrier analysis of Volume II reaches the conditional reduction

$$\mathfrak{e}_8 \supset \mathfrak{so}(10) \oplus \mathfrak{su}(4), \quad (2.1)$$

equivalently the $D_5 \oplus A_3$ maximal-rank branching. At the level of the adjoint representation,

$$\mathbf{248} = (\mathbf{45}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{15}) \oplus (\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}}) \oplus (\mathbf{10}, \mathbf{6}). \quad (2.2)$$

This is standard group theory, not a new particle prediction. Grand-unified embeddings of one Standard Model generation in the spinor $\mathbf{16}$ of $\text{Spin}(10)$, and its decomposition under $SU(5)$, have a long history (Georgi and Glashow, 1974; Fritzsch and Minkowski, 1975). Moreover, an E_8 representation by itself does not yield the observed chiral Standard Model; representation-theoretic obstructions must be addressed rather than hidden (Distler and Garibaldi, 2010).

The useful conventional identity is

$$\mathbf{16} \downarrow_{SU(5)} = \mathbf{10} \oplus \overline{\mathbf{5}} \oplus \mathbf{1}, \quad (2.3)$$

which is also realised by the even exterior algebra of five complex directions. Equation (2.3) packages one left-handed Weyl generation, including a sterile right-handed-neutrino conjugate.

2.2 Hypercharge and electric charge

Split the five complex directions into three colour directions and two weak directions. A generator constant on each block and traceless on the five-dimensional fundamental has, up to an overall normalization,

$$Y = \text{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right). \quad (2.4)$$

Exterior products add the diagonal charges, and electric charge is

$$Q = T_3 + Y. \quad (2.5)$$

The resulting multiplets agree with the conventional one-generation charge table (Navas et al., 2024).

TABLE 2.1. One generation written as left-handed Weyl fields.

Field	$SU(3)_c$	$SU(2)_L$	Y
Q_L	3	2	1/6
u_L^c	$\bar{\mathbf{3}}$	1	-2/3
d_L^c	$\bar{\mathbf{3}}$	1	1/3
L_L	1	2	-1/2
e_L^c	1	1	1
ν_L^c	1	1	0

ESTABLISHED RESULT 2.1 — CHARGE RECONSTRUCTION

Given the conventional $SU(5)$ block split and the normalization in Equation (2.4), the Standard Model hypercharges and electric charges follow. The fractional quark charges reflect tracelessness across three colour and two weak directions. This reconstruction does not explain why nature selects that embedding, why there are three families, or how symmetry breaking occurs.

The same normalization gives the familiar unified boundary value

$$\sin^2 \theta_W = \frac{3}{8}. \quad (2.6)$$

Equation (2.6) is a high-scale group-normalization relation, not the measured low-energy weak mixing angle. Renormalization-group evolution, thresholds, and the symmetry-breaking spectrum are required before comparison with laboratory data.

2.3 Anomaly cancellation is a necessary arithmetic check

For the fields in the table, the mixed and Abelian gauge-anomaly sums vanish:

$$[SU(3)]^2 U(1) : \quad 2\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \frac{1}{2}\left(-\frac{2}{3}\right) + \frac{1}{2}\left(\frac{1}{3}\right) = 0, \quad (2.7)$$

$$[SU(2)]^2 U(1) : \quad 3\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \frac{1}{2}\left(-\frac{1}{2}\right) = 0, \quad (2.8)$$

$$[\text{grav}]^2 U(1) : \quad 6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 = 0, \quad (2.9)$$

$$[U(1)]^3 : \quad 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1 = 0. \quad (2.10)$$

The purely non-Abelian local cubic anomaly vanishes for $SU(2)$, while the even number of weak doublets avoids the global $SU(2)$ anomaly. These cancellations validate the reproduced charge assignment. They do not independently derive it.

2.4 The proposed matter-action square

The corpus extends the gravitational square of Volume II from the geometry block to the full carrier. Its three algebraic weights are

$$W_L = u^2, \quad W_B = 2ur, \quad W_M = r^2, \quad W_L + W_B + W_M = 1, \quad (2.11)$$

where

$$r = \frac{1}{2\varphi} = 0.309016994\dots, \quad u = 1 - r. \quad (2.12)$$

It associates the three terms of the square with gauge, matter-coupling, and mass sectors. Internally,

$$\frac{W_M}{W_L} = \frac{1}{5} \quad (2.13)$$

is exact. Physically, however, an action is more than three coefficients: it requires fields, representations, covariant derivatives, kinetic terms, symmetry breaking, reality conditions, and equations of motion.

CONDITIONAL IDENTIFICATION 2.1 — THE MATTER SQUARE

If the carrier projection selects the conventional gauge and spinor modules and if the three terms of Equation (2.11) multiply correctly normalized gauge, fermion, and mass operators, then their relative coefficients are fixed by r . Those operators and their normalization have not yet been derived, so no Standard Model coupling or particle mass follows at present.

2.5 Chirality, mass, and the strong-CP boundary

The corpus proposes that reversible carrier directions support visible gauge transport, while an irreversible record condition protects chirality by excluding a traversal immediately composed with its reverse. This is a structural picture, not yet a quantum field theory. A finite decay group alone does not evade the representation-theoretic obstruction identified for four-dimensional E_8 unification (Distler and Garibaldi, 2010); a concrete chiral spectrum and anomaly-free action are required.

Likewise, identifying colour-block reversibility with a vanishing QCD vacuum angle does not calculate the strong-CP parameter. The standard strong-CP problem and its Peccei–Quinn/axion solution have well-defined field-theoretic content (Peccei and Quinn, 1977; Weinberg, 1978; Wilczek, 1978), and neutron electric-dipole searches constrain CP violation directly (Abel et al., 2020).

EXPOSURE 2.1 — THE DECISIVE MATTER CALCULATION

Construct a local, unitary, anomaly-free action; derive its chiral fermion content, gauge couplings, Yukawa matrices, symmetry-breaking pattern, and strong-CP sector; and run its parameters to measured scales. Charge-table reproduction and the coefficient ratio $1/5$ do not substitute for this calculation.

Hydrogen, finite designs, and atomic structure

3.1 The established hydrogen symmetry

The nonrelativistic Coulomb Hamiltonian

$$H = \frac{\mathbf{p}^2}{2\mu} - \frac{Ze^2}{4\pi\epsilon_0 r} \quad (3.1)$$

has, for bound states, the familiar energy spectrum

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2 \hbar^2 n^2}, \quad n = 1, 2, \dots \quad (3.2)$$

The angular momentum and Runge–Lenz vector generate an $SO(4)$ dynamical symmetry on a fixed negative-energy shell. Fock’s momentum-space construction maps the bound-state integral equation to a problem on S^3 (Fock, 1935). The momentum sphere and Equation (3.2) are established quantum mechanics; they cannot be counted as predictions of the carrier framework.

3.2 What the discrete polytopes actually establish

A finite set $X \subset S^3$ is a spherical t -design when averaging every polynomial of total degree at most t over X reproduces the spherical average (Delsarte et al., 1977). The 24-cell vertices form a 5-design, and the 120 vertices of the 600-cell form an 11-design (Coxeter, 1973; Conway and Sloane, 1999). Because a product of harmonics of degree at most ℓ has degree at most 2ℓ , a t -design exactly reproduces the corresponding inner products when

$$2\ell \leq t. \quad (3.3)$$

Hydrogenic states with principal quantum number n contain $\ell \leq n - 1$. Therefore

design	t	exact harmonic sector
24-cell	5	$\ell \leq 2 \quad (n \leq 3),$
600-cell	11	$\ell \leq 5 \quad (n \leq 6).$

(3.4)

INTERNAL RESULT 3.1 — FINITE QUADRATURE CUTOFFS

Equation (3.4) follows from the published design strengths and the degree of harmonic products. It validates exact finite quadrature for the stated sectors. It does not derive the Coulomb Hamiltonian, the Rydberg scale, relativistic fine structure, Lamb shifts, or transition rates.

The earlier corpus route that inferred the hydrogen spectrum from a carrier S^3 was circular: it used the same momentum-space symmetry structure that is obtained only after the Coulomb problem and its energy shell have been supplied. Self-duality of a lattice does not identify position and momentum spaces as one physical manifold without a symplectic/Fourier realisation. The scientifically retained result is finite sampling, not an independent derivation of hydrogen.

3.3 Atomic and chemical data as an observational atlas

The Atomic Web material records all 118 elements with identifiers, display data, and audit files. It is valuable as a provenance-controlled atlas and as an example of the Universopedia observational standard. The element classes themselves come from established nuclear charge, isotope, electronic-structure, and spectroscopy data; the colour web is a visualization, not a new periodic law.

EXPOSURE 3.1 — FROM QUADRATURE TO ATOMIC PHYSICS

A genuine carrier prediction must begin with a derived interaction and Hilbert space, then return at least one previously uninserted energy, degeneracy breaking, transition amplitude, or selection rule. Matching a known S^3 representation or sampling its harmonics is insufficient.

Classical theories as limits and translations

4.1 Electromagnetism

The established field theory is generated by

$$S_{\text{EM}} = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} \sqrt{-g} \, d^4x + \int A_\mu J^\mu \sqrt{-g} \, d^4x, \quad (4.1)$$

with $F = dA$, gauge redundancy $A \mapsto A + d\chi$, and charge conservation following from the source coupling and gauge symmetry. The corpus's Light/Boundary/Matter vocabulary can label propagation, connection, and retained source structure, but relabelling Maxwell theory is not a derivation. A new electromagnetic sector would need a distinct constitutive relation, coupling, dispersion law, or boundary prediction.

4.2 Thermodynamics and statistical mechanics

The corpus interprets entropy increase as growth of distinctions that cannot be harmlessly overwritten in a chosen record description. This is compatible with coarse-graining, but it must not be confused with the microscopic origin of thermodynamic irreversibility. For a distribution p_i ,

$$S = -k_B \sum_i p_i \ln p_i \quad (4.2)$$

is a state functional once the ensemble and coarse-graining are specified. Reversible microscopic equations, typicality, environmental interaction, initial conditions, and the selected macrovariables determine when a macroscopic entropy law follows.

Shannon entropy quantifies uncertainty in a coding distribution (Shannon, 1948); Landauer's bound concerns logically irreversible erasure implemented physically (Landauer, 1961; Bennett, 1982). Neither result proves a universal metaphysical prohibition on erasing records. The strongest corpus statement is target-relative: erasing a distinction that later changes an admissible readout makes the compressed state dynamically insufficient.

4.3 The status of the classical manuscripts

The legacy electromagnetism, thermodynamics, and statistical-mechanics texts are therefore retained as interpretive translations and consistency checks. Where they reproduce standard equations, the equations keep their standard evidential status and literature. Where they add only new names, no independent confirmation is claimed. Where they propose a new record-dependent constitutive law, that law enters the exposure ledger until an experiment is specified.

Cosmological baseline and algebraic dictionary

5.1 The standard background

For a spatially homogeneous and isotropic universe, the Friedmann equations are

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad (5.1)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \quad (5.2)$$

with component conservation

$$\dot{\rho}_i + 3H\left(\rho_i + \frac{p_i}{c^2}\right) = 0. \quad (5.3)$$

These equations follow from general relativity under the symmetry assumptions (Einstein, 1916; Friedmann, 1922). They define background expansion; they do not by themselves calculate recombination, the primordial spectrum, perturbation transfer, galaxy bias, or a dark-matter particle.

Planck’s six-parameter flat- Λ CDM analysis gives, among other values, $n_s = 0.9649 \pm 0.0042$, $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and matter density $\Omega_m \simeq 0.315$ for the stated data combination and model (Planck Collaboration, 2020b). The SHoES baseline distance ladder reported $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess et al., 2022). These are model- and pipeline-conditioned inferences, not direct readings of a single instrument dial.

5.2 The fixed partition

From the formal update in Volume I,

$$r = \frac{1}{2\varphi}, \quad u = 1 - r, \quad (5.4)$$

and the square of the two channels gives

$$W_L = u^2 = 0.477457514063 \dots, \quad (5.5)$$

$$W_B = 2ur = 0.427050983125 \dots, \quad (5.6)$$

$$W_M = r^2 = 0.095491502813 \dots \quad (5.7)$$

The partition and identities such as $W_L/W_M = 5$ are exact. The cosmological dictionary is a later hypothesis:

$$\Omega_b^{\text{UM}} = \frac{W_M}{2} = 0.0477457514 \dots, \quad (5.8)$$

$$\Omega_c^{\text{UM}} = \frac{W_B}{\varphi} = 4r^2u = 0.2639320225 \dots, \quad (5.9)$$

$$\Omega_{\text{DE}}^{\text{UM}} = 1 - \Omega_b^{\text{UM}} - \Omega_c^{\text{UM}} = 0.6883222261 \dots, \quad (5.10)$$

$$Y_{\text{He}}^{\text{UM}} = \frac{W_L}{2} = 0.2387287570 \dots, \quad (5.11)$$

$$N_{\text{eff}}^{\text{UM}} = 3 + \frac{W_M}{2} = 3.0477457514 \dots, \quad (5.12)$$

$$n_s^{\text{dict}} = 1 - W_M(1 - 2r) = 0.9635254916 \dots, \quad (5.13)$$

$$\tau^{\text{dict}} = 2r^3 = 0.0590169944 \dots \quad (5.14)$$

CONDITIONAL IDENTIFICATION 5.1 — COSMOLOGICAL DICTIONARY

Every number in Equation (5.14) is an exact algebraic output of the declared map. The map is not derived by the partition theorem, and several entries are standard-fit parameters with correlated likelihoods rather than independent observables. Agreement after the dictionary was proposed is retrospective and cannot be graded as a prediction.

5.3 The evolved readout and its selection cost

The Field Ledger later evolves the fixed pole, assigns a phase, uses helical turn count as position along the path, and weights readings by a resolution kernel. In compact form,

$$\mathcal{R}[O] = \frac{\sum_t H_5(N_t)O(C_t)}{\sum_t H_5(N_t)}, \quad N_t = \frac{360T_O}{\Delta\theta_t}, \quad H_5(N) = \frac{2s^{10}}{1 + s^{10}}, \quad s = \frac{N}{N+1}. \quad (5.15)$$

The implementation reproduces the reported table and moves all six selected residuals toward the chosen Planck comparison. That is a reproducible property of the algorithm. It is not an out-of-sample likelihood result: the observable set, channel typing, optical-depth path factor, and comparison target were already in contact with the data.

INTERNAL RESULT 5.1 — DETERMINISTIC RECONSTRUCTION

Given the declared evolution, phase assignment, turn count, and kernel in Equation (5.15), the corpus's six-entry table regenerates deterministically. Its mean absolute standardized residual decreases from 0.9979 to 0.6713. This is an in-sample reconstruction, not evidence with a calculable sampling significance.

The former “traversal payment”

$$P_T = 1 - \frac{E_1}{E_0} = 0.327356 \tag{5.16}$$

is consequently a score-improvement statistic for that typed audit. Its near-one-third value is not a universal constant, and the later typing ladder correctly reinterprets it as dependent on which records have been classified. Nothing about this score is frozen in the living edition.

The rate across cosmological scale

6.1 An exact mixture identity

Let $N = \ln a$ and define the total acceleration parameter

$$\epsilon = -\frac{d \ln H}{dN}. \quad (6.1)$$

For noninteracting components with constant equations of state w_i , write

$$\epsilon_i = \frac{3}{2}(1 + w_i), \quad \epsilon = \sum_i f_i \epsilon_i, \quad (6.2)$$

where $f_i = \rho_i/\rho$. Since

$$\frac{df_i}{dN} = 2f_i(\epsilon - \epsilon_i), \quad (6.3)$$

one obtains

$$\frac{d\epsilon}{dN} = 2(\epsilon^2 - \langle \epsilon_i^2 \rangle) = -2 \text{Var}_f(\epsilon_i) \leq 0. \quad (6.4)$$

If the Hamilton–Jacobi curvature parameter is defined by

$$\frac{d\epsilon}{dN} = 2\epsilon(\epsilon - \eta_H), \quad (6.5)$$

then, whenever $\epsilon \neq 0$,

$$\eta_H = \frac{\langle \epsilon_i^2 \rangle}{\langle \epsilon_i \rangle}. \quad (6.6)$$

Equation (6.5) is standard Hamilton–Jacobi flow (Salopek and Bond, 1990; Skenderis and Townsend, 2006); Equation (6.6) is its fluid-mixture rewriting under the assumptions just stated.

A cosmological constant has $w = -1$ and $\epsilon_\Lambda = 0$, so its direct contribution cancels from numerator and denominator of Equation (6.6). For pressureless matter plus Λ ,

$$\eta_H = \frac{3}{2}, \quad \epsilon(z) = \frac{3}{2} \frac{\Omega_m(1+z)^3}{\Omega_m(1+z)^3 + \Omega_\Lambda}. \quad (6.7)$$

ESTABLISHED RESULT 6.1 — WHAT CONSTANT η_H MEANS LATE

Equation (6.7) exactly rewrites the flat matter-plus- Λ background as the constant- $\eta_H = 3/2$ trajectory. This is a useful invariant representation of Λ CDM, not an independent derivation or confirmation of it. Radiation, curvature, interactions, and evolving equations of state modify the mixture.

6.2 Inflationary tilt and running

For single-field inflation in the Hubble slow-roll expansion, to first order,

$$n_s - 1 \simeq 2\eta_H - 4\epsilon, \quad r_{\text{tens}} \simeq 16\epsilon. \quad (6.8)$$

Here r_{tens} denotes the tensor-to-scalar ratio and must not be confused with the corpus retention fraction r . Planck found $n_s = 0.9649 \pm 0.0042$ and no evidence for running at its sensitivity (Planck Collaboration, 2020a); BICEP/Keck obtained $r_{0.05} < 0.036$ at 95% confidence from the 2018 observing-season analysis (BICEP/Keck Collaboration, 2021).

If η_H is treated as constant, differentiating the first-order relation gives, at the corresponding order,

$$\alpha_s \equiv \frac{dn_s}{d \ln k} \simeq -8\epsilon(\epsilon - \eta_H) \simeq -\frac{r_{\text{tens}}(1 - n_s)}{4}, \quad (6.9)$$

where $d \ln k = (1 - \epsilon)dN$ and higher-order slow-roll terms have been neglected. The inherited manuscript described Equation (6.9) too strongly. It is not exact: both the spectral-index relation and the conversion between N and $\ln k$ are approximations at this order.

FORECAST 6.1 — CONSTANT-CURVATURE INFLATION

Under single-field Hubble slow roll with approximately constant η_H , Equation (6.9) predicts negative running whose magnitude is below a few 10^{-4} when the quoted n_s and tensor bound are used. A robust positive running, or a magnitude incompatible with the complete higher-order constant- η_H calculation, would reject this condition.

6.3 The proposed retention map

The framework separately proposes that one carrier closure corresponds to one unit of logarithmic cosmological scale. Fixed retention would then give a constant exponent,

$$n_s^{\text{flow}} = 1 + \ln(1 - r^3) = 0.9700474 \dots \quad (6.10)$$

The arithmetic is exact, but the identification of a carrier closure with an e-fold or a mode-label interval is not derived. Equation (6.10) is a postdiction because n_s was known when the map was proposed.

EXPOSURE 6.1 — CLOSURE-TO-SCALE BRIDGE

Derive the map from carrier closure count to comoving wavenumber, including normalization and perturbation dynamics, without using the observed tilt. Until then, the closeness of Equation (6.10) to Planck is numerology with a clearly stated research target, not a primordial-spectrum calculation.

Dark matter, galaxies, and structure formation

7.1 Density identity versus physical substance

The exact internal identity

$$\Omega_c^{\text{UM}} = \frac{W_B}{\varphi} = 4r^2(1-r) = 0.2639320225 \dots \quad (7.1)$$

is numerically close to the cold-dark-matter density inferred in flat Λ CDM fits. It does not specify what clusters. A viable dark sector must supply a stress–energy tensor and, at minimum, the background and linear behavior

$$\dot{\rho}_c + 3H\rho_c = 0, \quad (7.2)$$

$$\ddot{\delta}_c + 2H\dot{\delta}_c - 4\pi G\rho_m\delta_m \simeq 0, \quad (7.3)$$

$$w_c \simeq 0, \quad c_s^2 \simeq 0, \quad Q_{\text{visible}} = 0. \quad (7.4)$$

The corpus proposes self-holding carrier modes that are neutral under the visible gauge projection. No mode operator, abundance calculation, perturbation closure, or nonlinear solution is yet available.

RESEARCH TARGET 7.1 — NEUTRAL SELF-HOLDING MODES

Derive a stable neutral spectrum whose stress–energy satisfies Equation (7.4); then predict linear transfer functions, halo profiles, substructure, lensing, and interactions from the same parameters. The numerical density identity is not admissible as a substitute for these outputs.

The earlier “vacuum crowding” mechanism is withdrawn. In the corpus’s own galactic estimate it is about eleven orders of magnitude too small and scales like vacuum density times a weak gravitational potential. The calculation remains in the provenance ledger precisely to prevent its accidental revival under a new label.

7.2 Simulation and galaxy ledgers

IllustrisTNG provides a public suite of magnetohydrodynamic cosmological simulations with subgrid galaxy-formation prescriptions (Springel et al., 2018; Nelson et al., 2019). SPARC provides 3.6- μm photometry and high-quality rotation curves for 175 disk galaxies (Lelli et al., 2016); these data exhibit a tight radial-acceleration relation between observed and baryonic accelerations (McGaugh et al., 2016). Both are valuable test surfaces, but they answer different questions.

The corpus’s TNG, density-gradient-flow, catalogue, planet-hunt, and age-matching directories are exploratory programmes and data-engineering records. They demonstrate pipelines, candidate filters, or possible signatures; they do not document a detection of a new planet, field, or dark-matter law. A post hoc fit to a galaxy catalogue is especially vulnerable to degeneracy among stellar mass-to-light ratio, gas distribution, distance, inclination, feedback, halo response, and selection.

7.3 A fair discriminating programme

A useful confrontation should proceed in this order:

1. derive the field or particle equations before selecting target galaxies;
2. calibrate nuisance parameters on a declared training set;
3. publish withheld-galaxy predictions for rotation, lensing, and where possible vertical dynamics;
4. compare against baryon-plus-halo and modified-dynamics baselines with the same nuisance freedom;
5. report failures, covariance, and selection functions as prominently as favorable cases.

EXPOSURE 7.1 — DARK-SECTOR DEFEAT CONDITION

If the derived neutral modes cannot reproduce the CMB transfer function, large-scale growth, galaxy and cluster lensing, and nonlinear structure with one consistent stress–energy model, the proposed dark-matter realisation fails even though Equation (7.1) remains an algebraic identity.

The microwave background and expansion history

8.1 What the existing forward solver computes

The corpus solver uses Saha ionization equilibrium to estimate recombination,

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_H} \left(\frac{m_e k_B T}{2\pi\hbar^2} \right)^{3/2} \exp\left(-\frac{\chi_H}{k_B T}\right), \quad (8.1)$$

then integrates the sound horizon and comoving angular-diameter distance,

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}, \quad c_s = \frac{c}{\sqrt{3(1 + R)}}, \quad quad R = \frac{3\rho_b}{4\rho_\gamma}. \quad (8.2)$$

With seven invariant readouts and the observed background temperature, it returns $H_0 = 66.965 \text{ km s}^{-1} \text{ Mpc}^{-1}$ after imposing its acoustic-angle condition. It also returns $z_* \simeq 1369$, far earlier than the realistic last-scattering value near 1090. The resulting acoustic peak locations are correspondingly 13–19% high.

INTERNAL RESULT 8.1 — LIMITED BACKGROUND SOLVER

The reported numbers reproduce within the Saha-based code. The program calculates background distances and approximate scales; it is not a CMB likelihood or Boltzmann solver. The common peak displacement is a useful diagnostic of inadequate recombination, not a successful spectrum prediction.

A scientifically sufficient implementation must evolve multilevel recombination and the photon, baryon, neutrino, metric, and proposed dark-sector perturbations, then output TT , TE , EE , lensing, and their joint likelihood. Borrowing a standard cold-dark-matter transfer function would test the dictionary but not the proposed matter mechanism.

8.2 The two-basin interpretation of H_0

Starting from the solver value $H_0^{\text{low}} = 66.965$, the corpus applies three closure conventions, including

$$H_0^{\text{high}} = H_0^{\text{low}}(1 - r^3)^{-3} = 73.261 \dots \quad (8.3)$$

Nearby variants give 72.893 and 73.168. The arithmetic is exact, and the transformed values are close to the SHoES baseline. But there is no derived observable map determining which experimental pipeline receives which closure count.

CONDITIONAL INTERPRETATION 8.1 — EXPANSION BASINS

If early- and late-universe distance indicators couple to distinct states of one scale-dependent field, then different inferred H_0 values need not estimate the same effective quantity. Equation (8.3) does not establish that premise. The basin-selection functional must be derived and applied prospectively to strong lenses, standard sirens, megamasers, red-giant distances, and inverse ladders.

8.3 Dark energy after DESI DR2

DESI DR2 baryon-acoustic-oscillation measurements are consistent with a flat Λ CDM description of the BAO data. When combined with CMB and supernova samples, the inferred preference for evolving dark energy depends on the supernova compilation and model combination (DESI Collaboration, 2025). It is therefore scientifically incorrect to describe evolution as a settled single-survey detection.

The corpus proposes that a fixed geometric vacuum record has exactly $w = -1$, while apparent evolution arises when readouts with different traversal counts are combined. This is a testable interpretation only after the traversal-to-likelihood map is specified. A physical equation of state cannot be dismissed merely because different supernova pipelines return different posterior contours; calibration and population systematics must be modelled quantitatively.

FORECAST 8.1 — STATIC-VACUUM PIPELINE TEST

Derive, before a later survey combination is examined, the predicted $H(z)$, $D_M(z)$, $f\sigma_8(z)$, lensing response, and pipeline-dependent bias for the proposed fixed-vacuum model. A statistically compelling evolving equation of state in a single appropriately modelled BAO/growth pipeline, incompatible with the derived bias, would reject the static-vacuum interpretation.

Vacuum scale, gravity tests, and the experimental ledger

9.1 The dimensionless vacuum readout

The corpus defines

$$\lambda_{\text{UM}} = \frac{4\pi}{\sqrt{3}} r^{240} = 2.860333 \times 10^{-122}, \quad (9.1)$$

using the 240 roots of E_8 , a triangular cell factor, and a genus-zero boundary factor. Equation (9.1) is an exact number after those choices. Identifying it with the observed dimensionless cosmological constant is highly convention-sensitive: the manuscript selects the non-reduced Planck length, and the reduced convention changes the dimensionless normalization by powers of 8π . The root-count exponent and geometric prefactor were also proposed after the vacuum hierarchy was known.

CONDITIONAL IDENTIFICATION 9.1 — VACUUM NUMBER

If each of 240 carrier crossings contributes one factor of r , if the observer boundary is genus zero, if the triangular cell and Gauss–Bonnet normalizations are physical, and if the non-reduced Planck convention is the correct observational dictionary, then Equation (9.1) maps to the vacuum term. The numerical match is a postdiction with several load-bearing conditions.

The associated dimensional construction

$$a_0 = \left(\frac{\hbar c}{\rho_\Lambda} \right)^{1/4} \simeq 88.1 \mu\text{m} \quad (9.2)$$

is the fourth-root length of a vacuum energy density. The relation is dimensional; interpreting a_0 or a weighted branch of it as a carrier-cell size is the new claim. A viable short-range-gravity forecast must specify the modified potential, coupling strength, composition dependence, and branch before comparison.

9.2 Equivalence and post-Newtonian gravity

The helical metric compression of Volume II uses

$$A = -\alpha_M, \quad B = \alpha_L + \frac{\nu}{3}. \quad (9.3)$$

Imposing $\nu = 3(\alpha_M - \alpha_L)$ makes the post-Newtonian slip

$$\gamma = \frac{\alpha_L + \nu/3}{\alpha_M} = 1 \quad (9.4)$$

identically. This proves that the chosen map cannot produce slip; it does not derive the map or its sources. Cassini constrained $\gamma - 1$ through radio tracking near solar conjunction (Bertotti et al., 2003), and MICROSCOPE found no weak-equivalence-principle violation at its final sensitivity (Touboul et al., 2022). Any helical field must pass these results together with laboratory, binary-pulsar, lensing, and gravitational-wave constraints.

The internal identity

$$(r\varphi)^2 = \frac{1}{4} \quad (9.5)$$

has the same coefficient as the Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4\ell_P^2)$ in natural units (Bekenstein, 1973; Hawking, 1975). Equality of coefficients is a contact, not a microstate count or a derivation of horizon thermodynamics.

9.3 A chronological defeat ledger

TABLE 9.1. Application programme: present result and next decisive test.

Sector		Present status	Decisive advance or defeat condition
Standard charges	Model	conventional $SU(5)/\text{Spin}(10)$ reconstruction	derive a local chiral action, running couplings, symmetry breaking, masses, and mixings
		Matter square exact coefficient identities, physical operator map open	equations of motion fail consistency, anomaly, or precision tests
Strong CP		reversibility interpretation only	calculate $\bar{\theta}$ and neutron EDM without inserting the observed bound

Sector	Present status	Decisive advance or defeat condition
Hydrogen	exact finite-design quadrature through stated harmonic sectors	predict an uninserted atomic energy or transition observable
Cosmological dictionary	exact algebra under a proposed map; retrospective comparison	withheld multi-observable likelihood rejects the common map
Constant η_H	exact fluid identity late; leading-order inflationary forecast	higher-order calculation conflicts with measured running/tensors
Dark matter	density identity plus unbuilt neutral-mode proposal	one stress-energy model fails CMB, growth, lensing, and halos jointly
CMB solver	background integrals with inadequate Saha recombination	full recombination and Boltzmann implementation fails peak phases, heights, damping, or lensing
Hubble basins	numerical transforms, selection rule absent	prospectively assigned intermediate probes do not occupy predicted basins
Static vacuum	conditional $w = -1$ interpretation	derived single-pipeline prediction is excluded by expansion and growth data
Vacuum number	convention-sensitive postdiction	physical normalization or pre-registered short-range signal fails
Helical gravity	algebraic $\gamma = 1$ under imposed source relation	derived source action violates equivalence, post-Newtonian, lensing, pulsar, or wave tests

9.4 What this volume establishes

The corpus makes several exact and useful contacts: the conventional one-generation charge pattern is reproduced from a five-direction block split; finite polytopes supply exact quadrature for specified hydrogenic harmonic sectors; the cosmological mixture admits the moment identity in Equation (6.6); and the software regenerates its declared algebraic and background readouts. None of

these closes the physical theory.

The remaining work is concentrated rather than diffuse. A matter action, a perturbation-complete cosmological solver, a dark-sector stress–energy tensor, a derived closure-to-scale map, and prospective measurement rules would turn the most important conditional identifications into genuine tests. Volume V records every source, correction, convention, certificate, and unresolved bridge so that those tests can be performed without freezing the present edition.

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